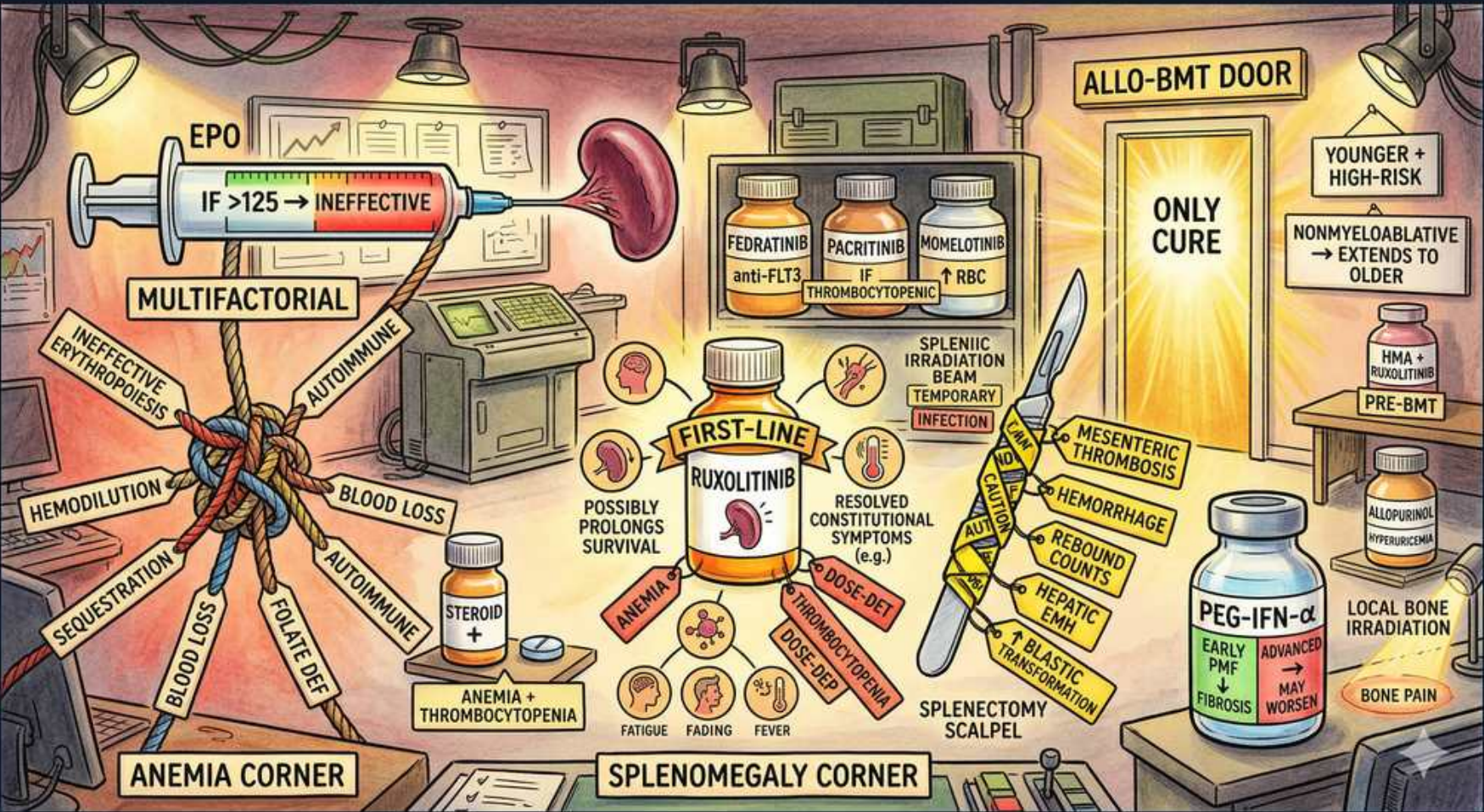


Polycythemia / MPN — Myelofibrosis treatment



1. Ruxolitinib first-line

WHAT YOU SEE

FIRST-LINE RUXOLITINIB bottle center

WHAT IT ENCODES

Ruxolitinib, an oral JAK1/JAK2 inhibitor, is first-line for intermediate-2/high-risk PMF with symptomatic splenomegaly and/or constitutional symptoms (COMFORT-I/II trials). Typical starting dose 15-20 mg PO BID, titrated by platelet count; it blocks the constitutively active JAK-STAT pathway regardless of whether the driver is JAK2, CALR, or MPL.

BOARD TRAP

WRONG: ruxolitinib only works in JAK2-mutated patients. It improves spleen size and symptoms across all driver genotypes (JAK2, CALR, MPL, and triple-negative) because it inhibits the shared downstream JAK kinases, not the mutation itself.

2. Rux shrinks spleen & helps Sx

WHAT YOU SEE

Resolved constitutional symptoms, fatigue/fever fading

WHAT IT ENCODES

Ruxolitinib produces durable spleen-volume reduction and dramatic relief of constitutional symptoms, with a modest overall-survival benefit. Crucially it is DISEASE-MODIFYING at best — it does not eradicate the clone or reliably reverse fibrosis, and abrupt discontinuation can trigger a cytokine-rebound 'withdrawal syndrome' (taper, do not stop suddenly).

BOARD TRAP

WRONG: ruxolitinib is curative and reverses marrow fibrosis. It is palliative/disease-controlling only — the ONLY curative option is allogeneic stem-cell transplant; rux shrinks spleen and eases symptoms but the clone and fibrosis persist.

3. Rux side effect: cytopenias

WHAT YOU SEE

ANEMIA + THROMBOCYTOPENIA dose-dependent icons

WHAT IT ENCODES

Because JAK2 signaling is required for normal erythropoiesis and thrombopoiesis, ruxolitinib's main on-target toxicities are dose-dependent ANEMIA and THROMBOCYTOPENIA, often worst in the first 8-12 weeks. It is also immunosuppressive (reactivation of HZV/zoster, TB, hepatitis B; opportunistic infection risk).

BOARD TRAP

WRONG: worsening anemia on ruxolitinib means the drug failed and should be stopped immediately. Early anemia/thrombocytopenia is an expected on-target effect managed by dose adjustment and transfusion — discontinuation also risks rebound; for anemia-predominant patients momelotinib is preferred.

4. Allo-BMT = only cure

WHAT YOU SEE

ALLO-BMT DOOR, ONLY CURE glowing

WHAT IT ENCODES

Allogeneic hematopoietic stem-cell transplant is the ONLY potentially curative therapy for PMF — it can eradicate the clone and reverse fibrosis. It carries high treatment-related mortality (graft-versus-host disease, infection), so it is reserved for higher-risk patients with acceptable performance status and a suitable donor.

BOARD TRAP

WRONG: every newly diagnosed PMF patient should proceed straight to transplant. Allo-HCT is restricted to intermediate-2/high-risk, transplant-eligible patients because of substantial mortality; low/intermediate-1-risk patients are managed with observation or rux, not transplant.

5. Transplant for younger/high-risk

WHAT YOU SEE

YOUNGER + HIGH-RISK, nonmyeloablative extends to older

WHAT IT ENCODES

Allo-HCT candidacy favors YOUNGER, fit patients with intermediate-2/high-risk disease (high DIPSS/DIPSS-Plus or high-molecular-risk mutations). Reduced-intensity/nonmyeloablative conditioning extends transplant to selected older patients (into the 60s-70s) who could not tolerate full myeloablation.

BOARD TRAP

WRONG: low-risk PMF patients benefit most from early transplant. Low-risk patients live long enough that upfront transplant mortality outweighs benefit — transplant is targeted to HIGH-risk disease where untreated survival is short.

6. Fedratinib (anti-FLT3)

WHAT YOU SEE

FEDRATINIB anti-FLT3 bottle

WHAT IT ENCODES

Fedratinib is a JAK2/FLT3 inhibitor approved for intermediate-2/high-risk MF, both first-line and after ruxolitinib failure. Dose ~400 mg PO daily. CRITICAL safety point: it carries a boxed warning for ENCEPHALOPATHY including Wernicke's — check and replete thiamine before and during therapy.

BOARD TRAP

WRONG: any MF patient who fails ruxolitinib has no remaining JAK-inhibitor options. Fedratinib (plus pacritinib and momelotinib) provides second-line JAK-inhibitor therapy after rux failure — just monitor thiamine to avoid Wernicke encephalopathy.

7. Pacritinib (thrombocytopenic)

WHAT YOU SEE

PACRITINIB THROMBOCYTOPENIC label

WHAT IT ENCODES

Pacritinib (JAK2/IRAK1/ACVR1 inhibitor, 200 mg PO BID) is the JAK inhibitor of choice for SEVERE THROMBOCYTOPENIA — it is approved for platelets <50,000/uL, a setting where ruxolitinib is contraindicated/poorly tolerated. Watch for diarrhea and QT prolongation.

BOARD TRAP

WRONG: ruxolitinib is the best choice when platelets are <50,000. Ruxolitinib is limited/contraindicated at very low platelet counts; PACRITINIB is specifically indicated for severe thrombocytopenia.

8. Momelotinib (anemia/RBC)

WHAT YOU SEE

MOMELOTINIB ↑RBC

WHAT IT ENCODES

Momelotinib (JAK1/JAK2 plus ACVR1/ALK2 inhibitor) uniquely IMPROVES anemia — by suppressing ACVR1 it lowers hepcidin, freeing iron for erythropoiesis. It is the preferred JAK inhibitor when ANEMIA predominates, controlling spleen/symptoms while raising hemoglobin and reducing transfusion need.

BOARD TRAP

WRONG: like other JAK inhibitors, momelotinib worsens anemia. It is the exception — via ACVR1/hepcidin suppression it IMPROVES anemia, making it the go-to agent for the anemic MF patient.

9. Splenectomy / irradiation

WHAT YOU SEE

SPLENECTOMY SCALPEL, SPLENIC IRRADIATION BEAM

WHAT IT ENCODES

Splenectomy or low-dose splenic irradiation is reserved for symptomatic, drug-refractory massive splenomegaly, transfusion-dependent anemia from sequestration, or severe portal hypertension. Both carry serious risk: perioperative bleeding, post-splenectomy infection, rebound thrombocytosis/leukocytosis, and hepatic/mesenteric/portal vein thrombosis; splenic irradiation gives only temporary relief and can cause severe cytopenias.

BOARD TRAP

WRONG: splenectomy cures the cytopenias and is a safe, definitive fix for the big spleen. It is palliative and risky — rebound extreme thrombocytosis/leukocytosis, fatal mesenteric/portal thrombosis, bleeding, and infection make it a last resort, not a cure.

10. Anemia: steroids/danazol

WHAT YOU SEE

ANEMIA CORNER, STEROID + RBC, knotted ineffective hematopoiesis

WHAT IT ENCODES

Anemia in PMF is multifactorial — ineffective erythropoiesis, marrow replacement by fibrosis, hemodilution from a huge spleen/plasma volume, splenic sequestration, hemolysis, and bleeding. Treatment options include corticosteroids, the attenuated androgen danazol, erythropoiesis-stimulating agents, immunomodulators (thalidomide/lenalidomide, especially with del(5q)), and RBC transfusion.

BOARD TRAP

WRONG: anemia in PMF is purely from marrow fibrosis, so only transfusion can help. It is multifactorial (sequestration, hemodilution, hemolysis, bleeding too); androgens/danazol, steroids, ESAs, and immunomodulators all have roles depending on mechanism.

11. EPO ineffective if >125

WHAT YOU SEE

EPO syringe IF >125 → INEFFECTIVE

WHAT IT ENCODES

Erythropoiesis-stimulating agents help only when endogenous EPO is LOW; if the serum EPO is already elevated (>125 U/L), exogenous EPO is unlikely to work and is not recommended. Always check the baseline EPO level before starting an ESA for MF-related anemia.

BOARD TRAP

WRONG: give an ESA empirically to every anemic MF patient. ESAs are futile when endogenous EPO is already high (>125 U/L) — measure EPO first to select responders.

12. PEG-IFN-alfa (early MF)

WHAT YOU SEE

PEG-IFN-α EARLY PMF, may worsen advanced

WHAT IT ENCODES

Pegylated interferon-alfa can be used in EARLY/lower-risk or hyperproliferative MF (and is favored in pregnancy and in PV/ET), with potential to reduce allele burden and induce molecular responses. In ADVANCED fibrotic disease it is poorly tolerated and may worsen cytopenias/fibrosis.

BOARD TRAP

WRONG: interferon is a good salvage option for advanced, heavily fibrotic high-risk MF. It is best in EARLY/proliferative disease; in advanced fibrosis it tends to worsen cytopenias and is not appropriate for high-risk salvage.

13. Bone pain (local irradiation)

WHAT YOU SEE

LOCAL BONE IRRADIATION, BONE PAIN

WHAT IT ENCODES

Localized low-dose external-beam radiation palliates focal bone pain from osteosclerosis/periostitis and treats symptomatic extramedullary hematopoiesis foci (e.g., paraspinal masses causing cord compression, EMH-driven pleural/pericardial effusions or ascites).

BOARD TRAP

WRONG: radiation in MF is given to the whole skeleton to treat the underlying disease. It is LOCAL palliation only — directed at a painful site or a symptomatic EMH focus — and never a disease-modifying or curative modality.